

Measures of Quality of Estimators

- How to measure the quality of different estimators?
- Suppose $f(x; \theta)$ for $\theta \in \Omega$ is the pdf (pmf) of a random variable X . A random sample $X_1, \dots, X_n$ is from the distribution of X . Consider a point estimator $Y_n = u(X_1, \dots, X_n)$ of θ .
 - Consistency: $Y_n \xrightarrow{P} \theta$, i.e., Y_n is close to θ for large sample size. MLE is consistent under suitable conditions.
 - Unbiasedness: $\mathbb{E}Y_n = \theta$. MLE may not be unbiased but they are usually asymptotically unbiased.
- If the two estimators of θ are unbiased, we would choose the one with smaller variance.

- For a given positive integer n , $Y = u(X_1, \dots, X_n)$ will be called a **minimum variance unbiased estimator**, (MVUE), of the parameter θ , if Y is unbiased, $\mathbb{E}Y = \theta$, and if the variance of Y is less than or equal to the variance of EVERY other unbiased estimator of θ .
- Example: Let $X_1, \dots, X_9$ denote a random sample from $N(\theta, \sigma^2)$. So, $\bar{X} = (X_1 + \dots + X_9)/9$ is $N(\theta, \sigma^2/9)$. $\bar{X}$ is an unbiased estimator of θ . The statistics X_1 is $N(\theta, \sigma^2)$, so X_1 is also an unbiased estimator of θ . $\text{var}(\bar{X}) < \text{var}(X_1)$ and we like $\bar{X}$. However we still cannot say that $\bar{X}$ is MVUE; that definition requires that the comparison be made with every unbiased estimator.

Decision Theory

- Let us now discuss the problem of a point estimation of a parameter from the DECISION THEORY standpoint.
- Let $Y = u(X_1, \dots, X_n)$ be a statistic on which we wish to base a point estimate of the parameter θ . Let $\delta(y)$ be that function of the observed value of the statistic Y which is the point estimate of θ . δ is called a **decision function** or a **decision rule**. One value of the decision function, say $\delta(y)$, is called a **decision**.

- With each pair $[\theta, \delta(y)]$, we will associate a nonnegative number $\mathcal{L}[\theta, \delta(y)]$ that reflects this seriousness. We call the function $\mathcal{L}$ the **loss function**.
e.g., $\mathcal{L}[\theta, \delta(y)] = (\theta - \delta(y))^2$.

- The expected (mean) value of the loss function is called the **risk function**.
If f_Y is the pdf of Y , the risk function $R(\theta, \delta)$ is given by

$$R(\theta, \delta) = \mathbb{E}\mathcal{L}[\theta, \delta(y)] = \int_{-\infty}^{\infty} \mathcal{L}[\theta, \delta(y)] f_Y(y; \theta) dy.$$

- It would be desirable to select a decision function that minimizes the risk $R(\theta, \delta)$ for all θ .
- But this is usually impossible because the decision function δ that minimizes $R(\theta, \delta)$ for one value of θ may not minimize $R(\theta, \delta)$ for another θ .

- Example: Let $X_1, \dots, X_{25}$ be a random sample from $N(\theta, 1)$. Let $Y = \bar{X}$ and let $\mathcal{L}[\theta, \delta(y)] = (\theta - \delta(y))^2$. We shall compare two decision functions $\delta_1(y) = y$ and $\delta(y) = 0$. The corresponding risk functions are

$$R(\theta, \delta_1) = \mathbb{E}[(\theta - Y)^2] = \frac{1}{25},$$

$$R(\theta, \delta_2) = \mathbb{E}[(\theta - 0)^2] = \theta^2.$$

If $\theta = 0$ then $R(0, \delta_2) = 0$. If θ differs from zero by very much, $\delta_2 = 0$ is a poor decision. For example, if $\theta = 2$, $R(2, \delta_2) = 4 > R(2, \delta_1) = 1/25$.

- If we had restricted our consideration to δ such that $\mathbb{E}\delta(Y) = \theta$, under this restriction, the risk function is the variance of the unbiased estimator $\delta(Y)$. Actually, $\bar{X}$ is the MVUE.

- Suppose we do not want to restrict ourselves to decision function δ such that $\mathbb{E}\delta(Y) = \theta$ for all θ . Instead we say that the decision function that minimizes the maximum of the risk function is the best decision function. $R(\theta, \delta_2) = \theta^2$ is unbounded, $\delta_2(y) = 0$ is not a good decision function.

$$\max_{\theta} R(\theta, \delta_1) = \max_{\theta} \frac{1}{25} = \frac{1}{25}.$$

$\delta_1(y) = \bar{x}$ seems to be a good decision since $1/25$ is small.

- Without some restrictions on the decision function, it is difficult to find the decision that has a risk function which is uniformly less than the risk function of another decision.
- **Minimax decision function:** If the decision function given by $\delta_0(y)$ is such that, for all θ ,

$$\max_{\theta} R[\theta, \delta_0(y)] \leq \max_{\theta} R[\theta, \delta(y)].$$

- squared error loss:

$$\mathcal{L}[\theta, \delta(y)] = (\theta - \delta(y))^2.$$

- absolute-error loss function:

$$\mathcal{L}[\theta, \delta(y)] = \begin{cases} 0, & |\theta - \delta(y)| \leq a \\ b, & |\theta - \delta(y)| > a \end{cases}$$

where a, b are two positive constants, is sometimes referred to as the **goal post loss function**

$$\begin{aligned}R(\theta, \delta) &= \mathbb{E}((\theta - \delta(Y))^2) \\&= \mathbb{E}[(\theta - \mathbb{E}\delta(Y) + \mathbb{E}\delta(Y) - \delta(Y))^2] \\&= (\mathbb{E}\delta(Y) - \theta)^2 + \text{Var}(\delta(Y)) \\&= \text{bias}^2 + \text{variance}\end{aligned}$$

- $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \theta)$, $\theta > 0$ and μ is unknown. Let $Y = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Consider the decision $\delta(y) = by$ and the squared error loss. Which b gives the smallest risk?

The Likelihood Principle

- Suppose two different sets of data from possibly two different random experiments lead to respective likelihood $L_1(\theta)$ and $L_2(\theta)$ that are proportional to each other. These two data sets provide the same information about the parameter θ and a statistician should obtain the same estimate of θ from either.

- Example: Suppose that two statisticians A and B observe 10 independent trials of a random experiment ending in success or failure. The probability of success is θ . Assume that each statistician observes one success in these 10 trials. However, that A had decided to take $n = 10$ such observations in advance and found only one success while B had decided to take as many observations as needed to get the first success which happened on the 10th trial.

An Example

The model of A is that $Y \sim \text{Bin}(10, \theta)$ and $y = 1$ is observed. B is considering that Z has geometric pdf $g(z) = (1 - \theta)^{z-1}\theta$, $z = 1, 2, \dots$ and $z = 10$ is observed. The relative frequency of success is

$$\frac{y}{n} = \frac{1}{z} = \frac{1}{10}.$$

$$\mathbb{E} \frac{Y}{10} = \frac{1}{10} \mathbb{E} Y = \theta$$

$$\mathbb{E} \frac{1}{Z} = \sum_{z=1}^{\infty} \frac{1}{z} (1 - \theta)^{z-1} \theta > \theta.$$

$1/Z$ is biased while $Y/10$ is unbiased. Should we adjust B's estimator so that it, too is unbiased?

$$L_1(\theta) = \binom{10}{y} \theta^y (1 - \theta)^{10-y}$$

$$L_2(\theta) = (1 - \theta)^{z-1} \theta$$

$$\hat{\theta} = \frac{1}{10}.$$

Note that $L_1(\theta) \propto L_2(\theta)$ and the likelihood principle states that statistician A and B should make the same inference. We would not adjust the second estimator to make it unbiased.